Prototype quantum neural networks under realistic noise: a reproduction, a calibrated noise study, a hardware plan, and a shot-noise programme

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Abstract

We reproduce the four experiments of Rosenhahn and Hirche, *Prototype Learning for Quantum Neural Networks* [1] — a five-bit counting task, the Iris flowers, a Flappy Bird controller and an ECG denoiser — and then ask what happens to each of them on a real quantum computer. Three of the four reproduce (the fourth, the Iris weight picture of their Fig. 9, does not); the numbers are in Table 1. We then build a noise layer from the published calibration data of seven quantum computers (IBM, Quantinuum, IonQ, Google, IQM, Rigetti), simulate every experiment under it exactly, and find that *gate noise at today's published levels does not break any of the small networks*: the bit-count histogram moves by a total-variation distance of 0.005–0.04, the Flappy controller still plays perfectly on all seven machines, and ECG denoising loses less than a decibel on the two machines we could afford to simulate. What does break them is the *finite number of measurements*: the Flappy controller's decisions sit a median of 0.019 from its flap threshold, so it needs about 10^4 measurements per frame, and the bit-count loss is biased by $31.8/N$ at N measurements. We therefore develop and test three remedies — margin-aware training, common random numbers for the paper's finite-difference optimiser, and a sequential measurement rule — and find that the sequential rule alone cuts the Flappy measurement budget by a factor of 4.6 at no loss of play, that margin-aware training does not change the margins that matter, and that common random numbers reduce the optimiser's derivative noise by 3–30× but cannot remove the bias of the plug-in loss. A compilation and cost study for each vendor closes with a concrete recommendation: run the seven-qubit Flappy controller's activation map on a trapped-ion machine first. Every number in this document is produced by a script from a recorded run directory; every calibration figure carries a source URL and a verbatim quotation; every choice the paper leaves open is marked as an assumption.

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1 Introduction

What the paper proposes. A quantum neural network is a small quantum circuit whose rotation angles are treated as trainable weights; the network's output is the probability of measuring one designated qubit in the state $|1\rangle$. Rosenhahn and Hirche [1] propose a particular building block, the *prototype neuron*, in which each input qubit is first rotated by a learned angle, then used to control a rotation of the neuron's own qubit, and finally rotated back (Fig. 1). The effect is that the neuron "fires" most strongly when the input pattern matches a stored prototype pattern, and the fact that the inputs are rotated back means the same inputs can be shared by many neurons and by many training examples *in superposition*. The paper demonstrates this on four tasks: learning a target histogram over all 32 five-bit strings at once (bit-count), classifying the Iris flowers, controlling a Flappy Bird game, and denoising an electrocardiogram (ECG). Training uses no gradient descent; it uses a finite-difference estimate of the slope inside a simulated-annealing loop (Section 3.2).

What we did. This document records three things. First, a **reproduction**: a clean-room implementation (no code from the authors), cross-checked against an independent simulator

written blind by a second agent, that regenerates every figure and table of the paper that can be regenerated from its text (Section 4). Second, a **noise study**: a description of the seven quantum computers for which vendors publish calibration data, as mathematical noise channels, and an exact simulation of each experiment under each of them (Section 5), plus a **compilation and cost study** of what it would take to run the circuits on each machine (Section 6). Third, a **shot-noise programme**: having found that the number of repeated measurements, not gate error, is what limits a hardware demonstration, we implement and test three remedies (Section 7).

How to read the numbers. Every quantitative statement below comes from a *run directory* — a folder under `experiments/runs/` holding the script's output together with the random seed, the commit hash of the code and the simulator backend that produced it. Run directories are named by time stamp (for example `bitcount_noise/20260820T145335Z`) and are quoted in the text; the figures are produced from them by one script (`experiments/report_figures.py`). Where the paper leaves a choice open (its optimiser step sizes, its train/test split, its game physics) we had to choose, and each such choice is listed in Appendix A. Every calibration number of a quantum computer carries, in the repository, the URL it was read from, the date, and a verbatim quotation (Appendix B).

Compute discipline. All simulations were run on one workstation (64 cores, one RTX 3090 GPU). No single run was allowed to exceed about two hours; two studies that would have (noise-aware training of the 11-qubit bit-count network, ≈ 13 hours for its grid; the full ECG/Iris noise grid, ≈ 8 hours) were deliberately *not* run and are reported as such, with their measured per-cell cost, rather than approximated.

Table 1: Scoreboard: what the paper reports and what we obtain. "Reproduced" means the qualitative claim and the quantitative value (within the precision the paper's figure allows) both come out; "discrepancy" means they do not, and the text explains what we tried.

Paper item	Paper value	Ours	Status
Fig. 3, activation surface	$(0,1)\to\max$, $(1,0)\to 0$	1.000 / 0.000 / 0.500 / 0.500 at the corners	reproduced
Fig. 5, target histogram	bar heights	(number of bins + 1)/112, all 32 bars within 0.0002	reproduced
Fig. 6, superposition training	$D_{\mathrm{KL}} \approx 0$ in ≈ 8 iterations	median < 0.005 at sweep 9; 5/6 seeds $< 10^{-3}$ at 20	reproduced
Fig. 6, iterative baseline	$D_{\mathrm{KL}} \approx 0$ by ≈ 120	stalls at ≈ 0.15 after 200 (our construction)	discrepancy
Fig. 7, Iris learning curves	classic overfits, prototype does not	classic test loss rises after 1–4 sweeps; prototype's falls monotonically	reproduced
Iris accuracy	$\approx 98\,\%$	95	in range

2 Notation and definitions

This section defines every symbol used later. A reader who knows quantum circuits can skim it; a reader who does not should find here everything needed to follow the rest.

Qubits, states, measurement. A *qubit* is a two-level quantum system with basis states $|0\rangle$ and $|1\rangle$. The state of n qubits is a unit vector $\psi \in \mathbb{C}^{2^n}$ whose components are indexed by n-bit strings; we write the string with *qubit 1 (the first input) as the most significant bit*, matching the paper's circuit figures, which number qubits from the top. Measuring all qubits returns the bit string x with probability $|\psi_x|^2$. When only some qubits are measured, the probability of their outcome is the sum of $|\psi_x|^2$ over the strings x that agree with it on those qubits (the *marginal*). A *mixed* state, needed as soon as noise enters, is a positive matrix ρ of unit trace (the *density matrix*); a pure state is $\rho = \psi\psi^\dagger$, and measurement probabilities are its diagonal entries.

Gates. A gate is a unitary matrix acting on one or two qubits. The ones the paper uses are

$$R_X(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad H = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \tag{1}$$

$$CR_X(\theta) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_X(\theta) \quad \text{(rotate the target by } \theta \text{ only if the control is } |1\rangle\text{)}. \tag{2}$$

$R_X(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle - i\sin\frac{\theta}{2}|1\rangle$, so after $R_X(\theta)$ a qubit prepared in $|0\rangle$ is measured as 1 with probability $\sin^2(\theta/2)$. H turns $|0\rangle$ into the equal superposition $(|0\rangle + |1\rangle)/\sqrt{2}$; H on every input qubit prepares all 2^n input strings at once with equal weight.

Encoding a real number as an angle. A feature value $s \in [0, 1]$ enters the circuit as $R_X(\theta(s))$ on its input qubit. The paper does not state $\theta(s)$; its companion paper [2] (Fig. 4) prints $CR_X(1.369)$ for the value 0.4 and $CR_X(1.7722)$ for 0.6, which pins

$$\theta(s) = 2\arcsin\sqrt{s}, \qquad \text{so that } P(\text{qubit reads } 1) = s. \tag{3}$$

We use Eq. (3) throughout and report the alternative $\theta = \pi s$ as an ablation where it matters (ECG, Section 4.4).

Networks and their output. A network has n_{in} input qubits, n_{hid} hidden qubits (one per hidden neuron) and one output qubit; $n = n_{\text{in}} + n_{\text{hid}} + 1$. Its output for a given input is the *activation*

$$P(\text{out} = 1) \;=\; \Pr[\text{output qubit reads } 1], \tag{4}$$

a number in $[0, 1]$. For a classifier it is thresholded (Flappy: flap iff $P(\text{out} = 1) > \frac{1}{2}$); for a regressor it is the predicted value (ECG: the denoised sample).

The superposition observable and the training loss. For the bit-count and Iris tasks the inputs are prepared in superposition and the measured joint distribution of (input string x, output bit) is read. The *model distribution* is

$$q(x) \;=\; \frac{P(\text{inputs} = x,\ \text{out} = 1)}{\sum_{x'} P(\text{inputs} = x',\ \text{out} = 1)}, \tag{5}$$

i.e. the input distribution *conditioned on the output having fired* — the output amplitude acts as a weight on each input pattern (paper, Fig. 4 caption). It is compared to a target distribution $p(x)$ by the Kullback–Leibler divergence (paper, Eq. (4)),

$$D_{\text{KL}}(p \,\|\, q) \;=\; \sum_x p(x)\,\log\frac{p(x)}{q(x)} \;\geq 0, \qquad D_{\text{KL}} = 0 \iff p = q, \tag{6}$$

with $0 \log 0 = 0$ and q floored at 10^{-12} so the value stays finite (our choice; the paper does not say). A second dissimilarity we use for noise is the *total-variation distance* $\mathrm{TV}(p, q) = \frac{1}{2} \sum_x |p(x) - q(x)| \in [0, 1]$, the largest difference in probability that any event can have under the two distributions.

Noise channels. Real gates are not unitary. A *channel* is a linear map $\rho \mapsto \sum_i K_i \rho K_i^\dagger$ with $\sum_i K_i^\dagger K_i = I$ (the K_i are its *Kraus operators*); it sends density matrices to density matrices. Three channels describe a calibrated machine:

- *Depolarising* with probability p: with probability $1 - p$ nothing happens, with probability p the state of the affected qubit(s) is replaced by the maximally mixed state. A vendor's "average gate error" ε is converted to p by $\varepsilon = p\,(1 - 2^{-k})\,\frac{2^k}{2^k+1}$ for a k-qubit gate ($p = 2\varepsilon$ for one qubit, $p = \frac{4}{3} \cdot \frac{5}{4}\varepsilon$ for two), which makes the channel's average fidelity equal to $1 - \varepsilon$ (cross-checked against Qiskit Aer).
- *Relaxation* over a time t with the vendor's T_1 (energy decay time) and T_2 (phase decay time): $|1\rangle$ decays toward $|0\rangle$ with probability $1 - e^{-t/T_1}$ and superpositions lose coherence by e^{-t/T_2}. It is applied during every gate (for its duration) and during every idle interval, which we schedule as early as possible, exactly as IBM's compiler does.
- *Readout confusion*: the recorded bit differs from the true one with probability p_{01} (true 0 read as 1) or p_{10} (true 1 read as 0).

The resulting *noise model* of a machine is the rule "after each gate, apply the gate's depolarising channel and its relaxation; on every idle qubit, apply relaxation for the idle time; at the end, apply readout confusion". Its parameters come from the vendor data of Appendix B; a *noise scale* s multiplies all of p, $1/T_1$, $1/T_2$ and the readout errors together ($s = 1$ is the published machine, $s = 0$ is noiseless).

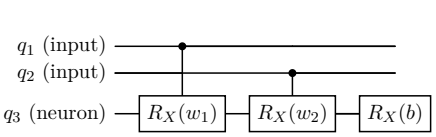
Shots. A quantum computer does not return probabilities; it returns one bit string per run, and probabilities are estimated by repeating the circuit N times (N *shots*). An estimate $\hat{p} = k/N$ of a probability p has standard deviation $\sqrt{p(1-p)/N} \le 1/(2\sqrt{N})$. We call $\delta = |P(\text{out} = 1) - \frac{1}{2}|$ the *decision margin* of a thresholded output: a frame decided from N shots is decided wrongly with probability $\approx \Phi(-2\delta\sqrt{N})$, where Φ is the standard normal distribution function.

Fidelity. For a compiled circuit on a given machine we quote the estimate

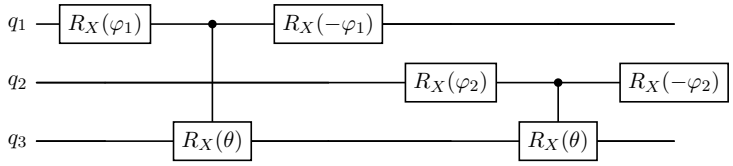
$$F \;\approx\; \prod_{\text{2q gates}} (1 - \varepsilon_2) \prod_{\text{1q gates}} (1 - \varepsilon_1) \prod_{\text{qubits}} F_{\text{idle}}(t) \prod_{\text{qubits}} \left(1 - \tfrac{p_{01}+p_{10}}{2}\right), \tag{7}$$

the probability that no error at all occurred in the run, with $F_{\text{idle}}(t) = \frac{1}{2} + \frac{1}{6}(2e^{-t/T_2} + e^{-t/T_1})$ the average fidelity of relaxation over the circuit's duration t [7]. It is a first-order bookkeeping estimate, not a simulation; the simulations are reported next to it.

Abbreviations. KW–SA: the paper's optimiser (Section 3.2). PSNR: peak signal-to-noise ratio, $10 \log_{10}(R^2/\mathrm{MSE})$ with R the signal range and MSE the mean squared error, in decibels (dB). SSIM: structural similarity index [11], a number in $[0, 1]$ comparing local means, variances and covariance of two signals. SWAP: the gate that exchanges two qubits, inserted by a compiler when a two-qubit gate is requested between qubits that are not physically connected; it costs three native two-qubit gates.



(a) Classic neuron (paper, Fig. 1): one controlled rotation per input and a bias rotation on the neuron's qubit. The inputs are untouched.



(b) Prototype neuron (paper, Fig. 2): each input is rotated by a learned φ_i, controls a rotation by the *same* θ of the neuron's qubit, and is rotated back.

Figure 1: The two neuron types. A network (paper, Figs. 4, 5, 10, 12) is a layer of n_{hid} hidden neurons, each touching all n_{in} inputs, followed by one classic output neuron controlled by the hidden qubits. Our implementation (`ptqnn.circuits.neurons`) builds exactly these gate lists; the un-computation $R_X(-\varphi_i)$ is what lets several prototype neurons share the same inputs and what lets a dataset be presented in superposition.

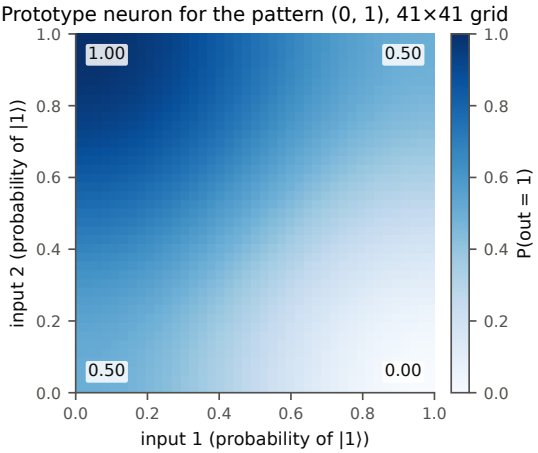


Figure 2: Activation $P(\text{out} = 1)$ of one prototype neuron with prototype $(0, 1)$ as a function of the two inputs (each given as the probability that its qubit reads 1, encoded by Eq. (3)). The corner values are exact. Run `bitcount/20260820T135930Z`.

3 The models and the optimiser

3.1 Classic and prototype neurons

Figure 1 shows the two neuron types as circuits. The *classic* neuron is the quantum perceptron of [2]: its qubit receives a bias rotation $R_X(b)$ and one controlled rotation $CR_X(w_i)$ from each input; the inputs are left unchanged. The *prototype* neuron rotates input i by $R_X(\varphi_i)$, lets it control a rotation by a shared angle θ of the neuron qubit, and rotates it back by $R_X(-\varphi_i)$. The pair $(\varphi_1, \ldots, \varphi_{n_{\text{in}}})$ is the stored *prototype*: when the input state is the one that $R_X(-\varphi_i)$ would map to $|1\rangle$ on every qubit, every controlled rotation fires and the neuron's activation is maximal; when no input matches, none fires and the activation is zero (Fig. 2). A network uses prototype (or, for comparison, classic) hidden neurons and always a classic output neuron. Parameter counts: the 11-qubit bit-count network (5 inputs, 5 hidden) has $5 \times (5 + 1) + (5 + 1) = 36$ angles; the 7-qubit Flappy network (2 inputs, 4 hidden) has $4 \times 3 + 5 = 17$.

A sanity check that a reader can verify by hand: a single prototype neuron with two inputs, prototype $(\pi, 0)$ and $\theta = \pi/2$, fed the basis input $(0, 1)$, gives $P(\text{out} = 1) = 1$; fed $(1, 0)$ it gives 0; fed $(0, 0)$ or $(1, 1)$ it gives $\frac{1}{2}$. Figure 2 is this neuron's activation over all real-valued inputs — the paper's Fig. 3. The paper's own surface bottoms out near 0.2 rather than 0, because its parameters (not given) differ from these.

3.2 The optimiser: finite differences inside simulated annealing

The paper trains with no gradient descent. It estimates the slope of the loss f along one parameter by the *central difference* $(f(x + c\,e_i) - f(x - c\,e_i))/2c$ (Kiefer–Wolfowitz [3]), proposes a step against it, and accepts or rejects the proposal by the *Metropolis rule* of simulated annealing: always accept a decrease of the loss, accept an increase Δ with probability $e^{-\Delta/kT}$ at temperature T (paper, Eq. (3)), cooling T geometrically. Neither paper gives the step sizes a_n and c_n, the temperature schedule or what one "iteration" is, so the following is our choice (listed in Appendix A): $a_n = a_0/(n+1)^{0.602}$, $c_n = c_0/(n+1)^{0.101}$ (Spall's standard exponents [4]); one *sweep* visits every parameter once in random order, evaluates the two probes $x \pm c_n e_i$ and the proposed step, and puts the best of the three to the Metropolis rule. One sweep therefore costs $3 \times$ (number of parameters) circuit evaluations — 108 for the bit-count network. Whole-vector steps (one gradient estimate of all parameters, then one step) stalled near $D_{\mathrm{KL}} \approx 10^{-2}$ at every setting tried, so the coordinate sweep is the default; it is what Fig. 4 shows. For a rotating mini-batch of training episodes (Flappy) the incumbent loss is re-evaluated every sweep, because comparing a candidate scored on one batch against an incumbent scored on another is meaningless (Section 4.3).

3.3 Simulators

The reference simulator is plain NumPy (state vectors for noiseless runs, density matrices for noisy ones) and is the oracle for everything else: an independent density-matrix simulator written blind by a second agent agrees with it to 10^{-12} on random circuits; a C density-matrix library (`orkan`), a Monte-Carlo trajectory simulator for circuits too large for density matrices (Iris at 17 qubits), a GPU density-matrix backend, and one shared batched state-vector kernel used by all task-specific fast paths are each tested against it on *full output distributions*, never only on $P(\mathrm{out} = 1)$. The noise model's idle scheduling reproduces Qiskit's as-soon-as-possible scheduler delay for delay, and its channels agree with Qiskit Aer's on a noisy three-qubit circuit to 2×10^{-15}.

4 Reproduction of the four experiments

4.1 Bit-count: learning a histogram over all inputs at once

Task. Five input qubits are placed in equal superposition of all 32 strings; the network is trained so that the model distribution Eq. (5) matches a target histogram that rises with the number of ones in the string. The paper gives the target only as a bar chart (Fig. 5, bottom right); reading the bars gives $p(x) = (\text{number of ones in } x + 1)/112$ for every one of the 32 bars to within 0.0002 ($112 = \sum_x (\text{ones} + 1)$ is the normalisation). The network is the paper's Fig. 5: 5 inputs, 5 prototype hidden neurons, one classic output, 11 qubits.

Result. Figure 3 shows the target and the trained model: $D_{\mathrm{KL}} = 5.5 \times 10^{-4}$. (The paper's own printed model histogram, read off the figure, has $D_{\mathrm{KL}} \approx 2.6 \times 10^{-3}$ against the same target.) Figure 4 (left) is the paper's Fig. 6: from a random start the median of six optimiser seeds crosses $D_{\mathrm{KL}} = 0.005$ at sweep 9 and five of six seeds are below 10^{-3} at sweep 20; one seed stalls at 0.014. The paper reports "≈ 0 in about 8 iterations" without defining an iteration; against our sweeps it matches.

The iterative baseline does not reproduce. The paper's Fig. 6 also shows a slower "iterative" curve, training on one input at a time and reaching ≈ 0 by about 120 iterations. The paper describes this only as "optimizing examples iteratively"; the companion paper [2] describes sample-

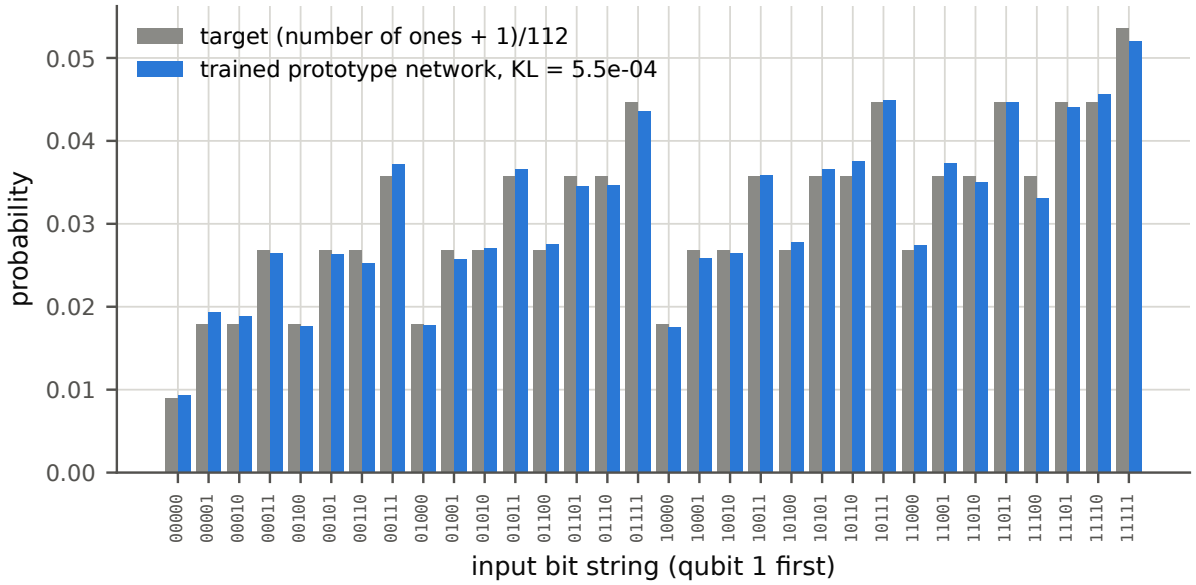


Figure 3: Paper Fig. 5. Target histogram (grey) and the trained prototype network's model distribution (blue) over the 32 input strings. Run `bitcount_noise/20260820T145335Z` (seed 1 of the bit-count training).

wise squared-error updates with the same optimiser but gives no schedule. Our construction — one optimiser sweep on the squared error of one randomly chosen input per iteration — stalls at a median $D_{\mathrm{KL}} \approx 0.15$ after 200 iterations (Fig. 4, right), and a sweep over step sizes $a_0 \in \{2, 5, 20, 50, 100\}$, two decay exponents, both update modes and a scaled per-sample target left the median between 0.05 and 0.3 (best single run 0.046). Updating on one input at a time thrashes between inputs. We record this as a discrepancy: whatever the authors' iterative scheme was, ours is not it.

4.2 Iris: a dataset in superposition

Task and encoding. Each of the four Iris features is quantised into three clusters by k-means, giving a 12-bit one-hot code per flower; the whole training set is loaded as one superposition state over its distinct codes, and the label is encoded in the amplitude. The paper does not state how the label enters; reading its Fig. 8 bar heights ($1/86, 4/86, 9/86$ on the training set and $1/154, 4/154, 9/154$ on the test set, two independent integer normalisations) fixes the target weight of class $\ell \in \{0, 1, 2\}$ to be proportional to $(\ell+1)^2$ — the amplitude carries $\ell+1$ and the histogram its square. No other exponent produces both normalisations. The paper also does not state the train/test split; we use a stratified one-third training split (50 flowers) and report the sensitivity. The network has 12 inputs, 4 hidden neurons and one output (17 qubits); a classic-neuron network of the same size is trained for comparison. Predictions are made by reading each code's weight and rounding to the nearest label.

Result. Figure 5 is the paper's Fig. 7 on three data splits: the classic network drives its training loss to 10^{-5} within a few sweeps while its test loss bottoms out early and stays at 0.03–0.04 (it memorises); the prototype network descends more slowly on both and its test loss is still falling at sweep 40. Test accuracy: 95 %, 91 % and 98 % on the three splits for the prototype network against a paper value of "≈ 98 %"; the encoding itself caps accuracy at 97 %, 99 % and 99 % on those splits, because two flowers of different species can share a 12-bit code. Figure 6 is Fig. 8 for the test split. The Fig. 9 picture — a two-dimensional embedding in which the prototype weights are claimed to "span the data" — does not reproduce: both weight sets cluster away from the

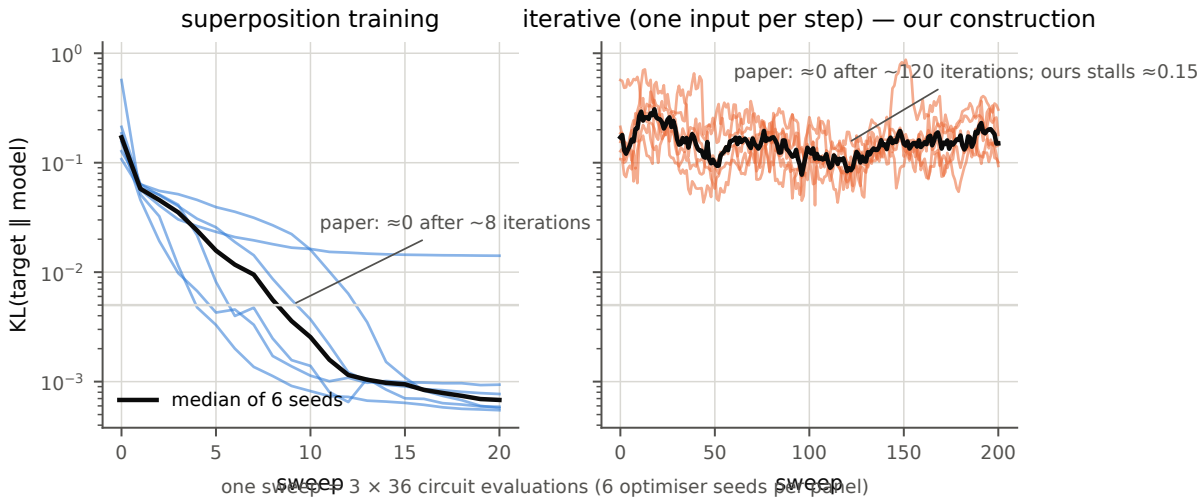


Figure 4: Paper Fig. 6. Left: superposition training, six seeds and their median. Right: our construction of the iterative baseline, which does not reach the paper's curve. Run `bitcount/20260820T135930Z`.

data in our embedding; we do not know what the authors plotted.

On the split. Fig. 8 shows 15 non-empty bars for the training set and 30 for the test set. Our 50-flower training splits produce 14, 12 and 14 distinct codes (close to 15), but the 100-flower test splits produce only 20, 20 and 18 (not 30): a one-hot code with three clusters per feature does not spread 100 flowers over 30 codes on any split we tried. The figure therefore supports the *direction* of the split (test larger than train) but not its ratio, and the ratio remains our assumption.

4.3 Flappy Bird: a seven-qubit controller

Task. Two features — the bird's height and the centre of the next gap, both scaled to $[0, 1]$ — enter two input qubits by Eq. (3); four prototype hidden neurons and one classic output decide "flap" when $P(\text{out} = 1) > \frac{1}{2}$ (paper, Fig. 10). The paper says the trained network "plays the game perfectly" and that a run was stopped after a score of 450 000 that took six days; it gives no reward, no training protocol and no game constants. We built a deterministic, window-free clone of the game. Its physics are our choice: the classic constants (flap impulse -9, gap 100, pipe spacing 144) are provably unplayable by a memoryless policy on two features (one flap overshoots by 40.5 px against a 38 px budget), so we use impulse -7, gap 120, spacing 180. Training minimises the negative episode return (0.01 per frame survived plus 1 per pipe) plus a small tracking term $|\text{height} - \text{gap}|$ that gives the optimiser a slope in the region where every policy dies within 30 frames. 300 random parameter vectors are scored on short episodes, the best seeds 25 sweeps of the optimiser on five 1500-frame training episodes.

Result. The trained network passes 110 pipes in 5 000 frames — the cap, i.e. a perfect game — on all 20 held-out game seeds, for all 8 optimiser seeds tried, in about three minutes of training (run `flappy/20260820T143753Z`). Figure 7 shows what it learned: a curved decision boundary in the (height, gap) plane, flapping when the bird is below the gap. A finding about the optimiser: rotating the five training episodes among a pool of ten every sweep gave 0/25 perfect games in the first session, and was recorded as a negative result; the cause was that the optimiser kept its incumbent's loss from the previous batch, so every Metropolis comparison compared scores on different games. Re-scoring the incumbent on each new batch restores 25/25 perfect games (runs `flappy/batch0`, `flappy/batch5`), so the rotation neither hurts nor helps here.

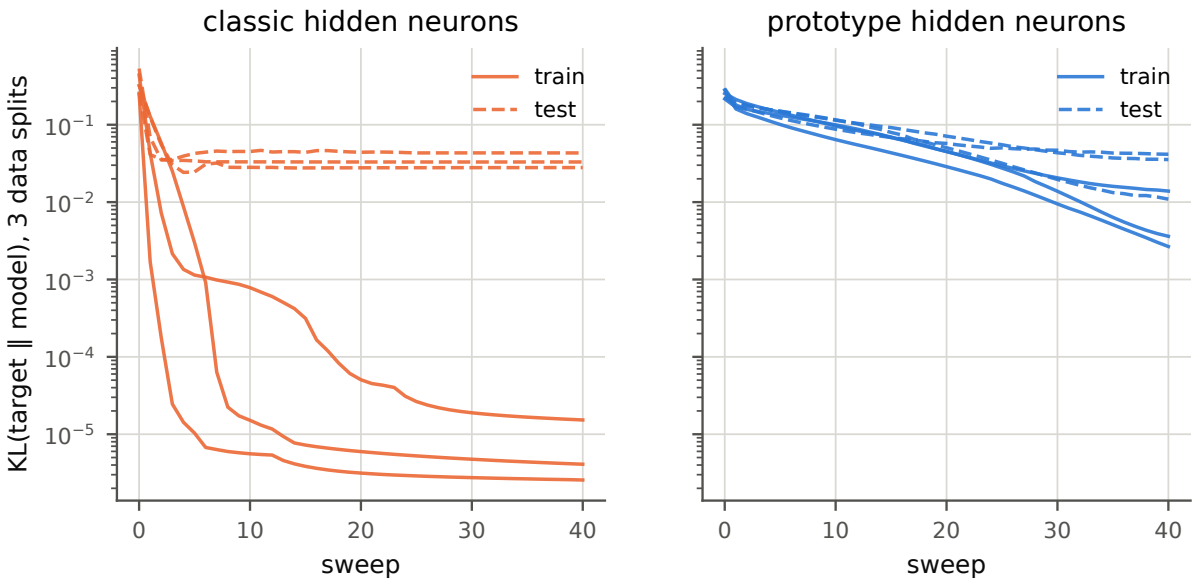


Figure 5: Paper Fig. 7. Training (solid) and test (dashed) loss per sweep for three data splits. Runs `iris/20260820T144849Z`, `...T145454Z`, `...T145455Z`.

4.4 ECG denoising

Task. A window of five consecutive noisy samples $s(t-2), \ldots, s(t+2)$ enters five input qubits; one to three prototype hidden neurons and one output predict the clean centre sample as $P(\text{out} = 1)$ (paper, Fig. 12). The paper does not say which recording it used; its Fig. 13 shows a smooth two-beat trace of 1 800 samples in $[0, 1]$ with additive noise of standard deviation ≈ 0.03. We generate a synthetic ECG of the McSharry form [12] tuned to that figure, add Gaussian noise of $\sigma = 0.02$ (noisy input 32.0 dB), train on the first half and test on the second, and repeat on PhysioNet MIT-BIH record 100 for comparison. Quality is measured by PSNR and SSIM (paper, Eqs. (6)–(8)).

Result. Figure 8 is the paper's Fig. 13; Fig. 9 is its Table I. We obtain 37.4, 38.0 and 38.1 dB for one, two and three prototype neurons against the paper's 32.9, 40.00 and 40.04: within about 2 dB at two and three neurons, and 4 dB *above* the paper at one neuron. The prototype network beats the classic one at one and two neurons and on every SSIM; the classic one wins at three. Two notes. The numbers cannot match exactly because the signal is not the paper's; and the alternative angle encoding $\theta = \pi s$ costs 0.6–7 dB on this task, which is independent evidence for Eq. (3). On MIT-BIH record 100 (run `ecg/20260820T165912`) the same networks reach 36–38 dB.

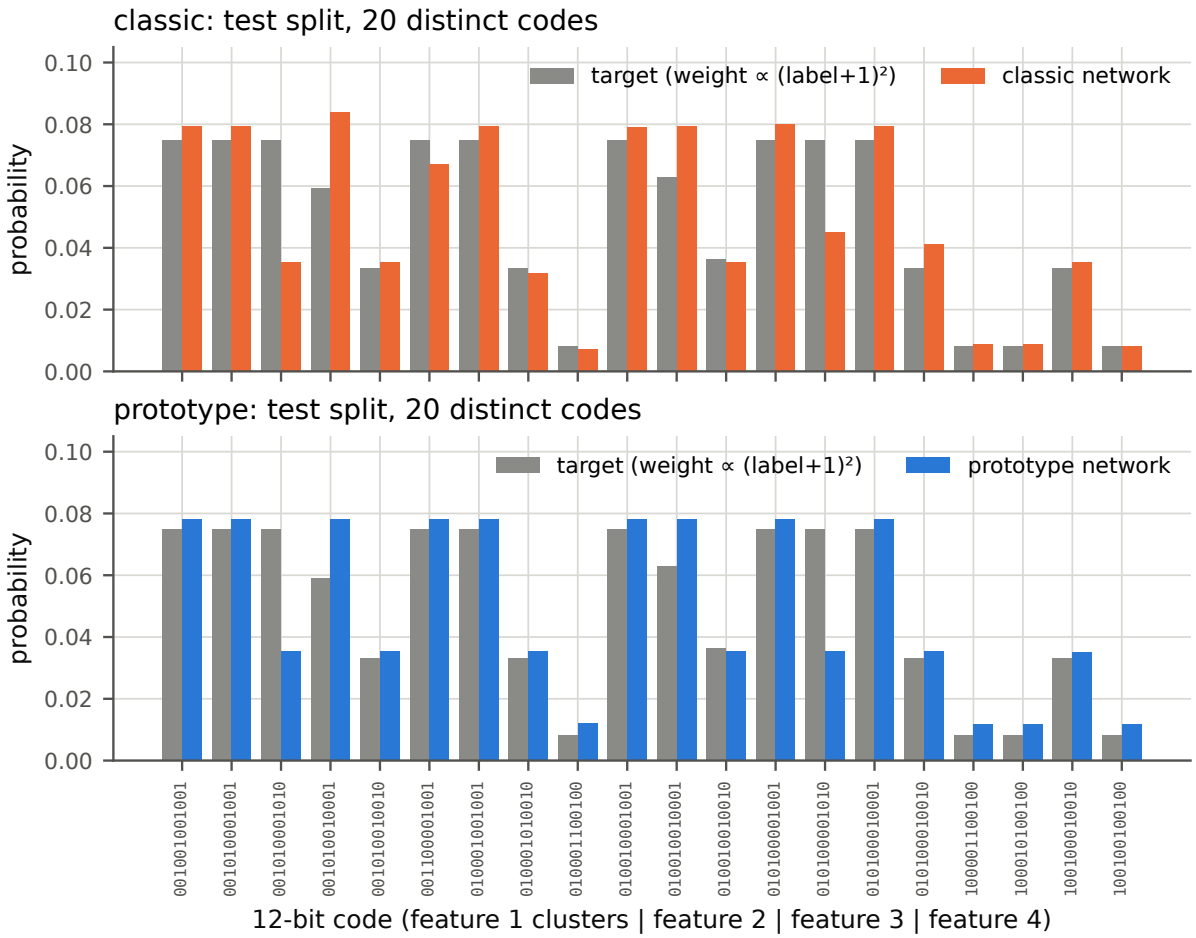


Figure 6: Paper Fig. 8, test split of seed 0: target weight per distinct code (grey) against the trained network's weight, classic (top) and prototype (bottom). The bars are the probabilities of reading each code together with the output firing, renormalised.

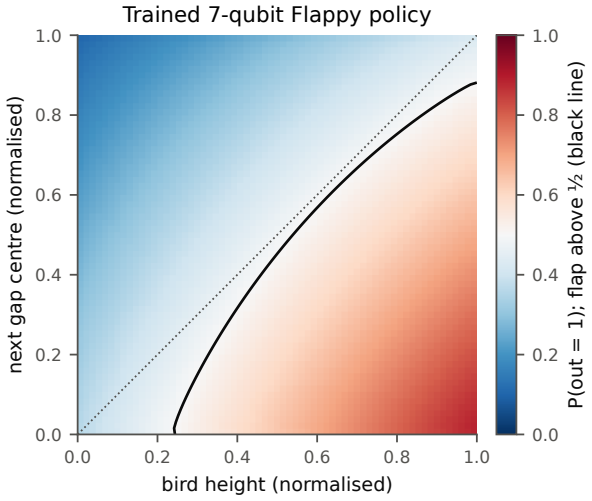


Figure 7: The trained Flappy controller's activation $P(\text{out} = 1)$ over the two observations; the black line is the decision boundary $P(\text{out} = 1) = \frac{1}{2}$ (flap to its right), the dotted diagonal is "bird at gap height". Run `flappy/20260820T143753Z`.

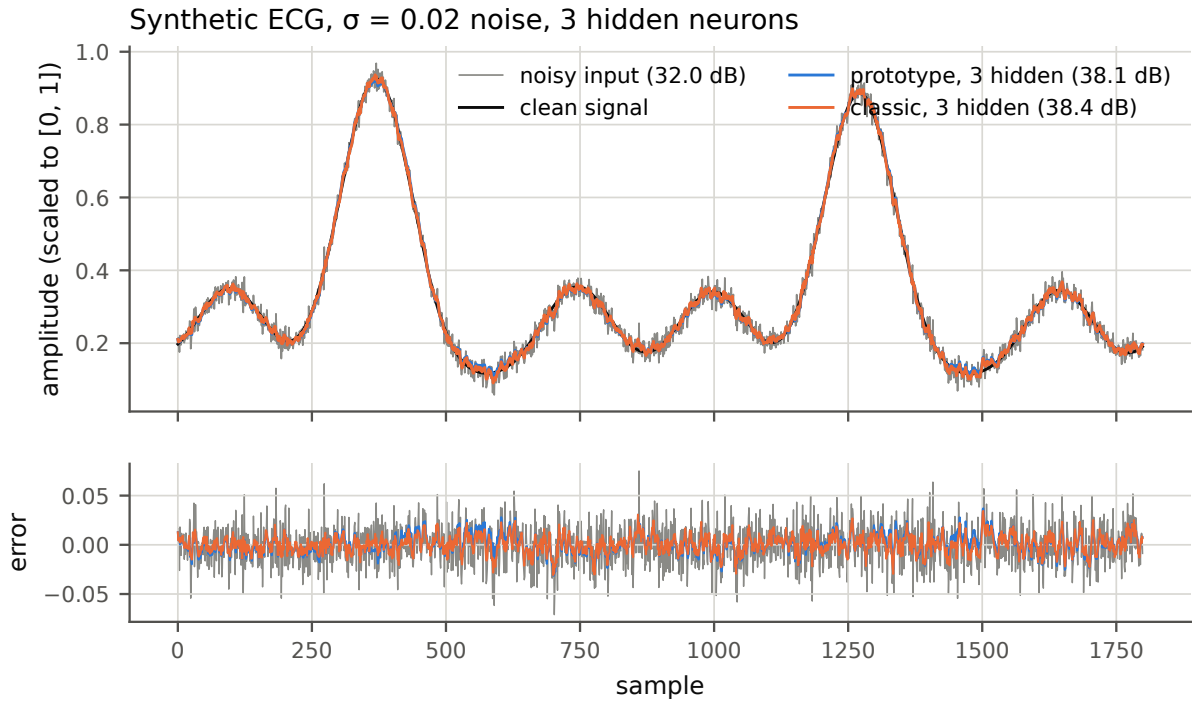


Figure 8: Paper Fig. 13. Noisy input, clean signal and the two three-neuron networks' outputs (top); their errors (bottom). Run `ecg/20260820T162014`.

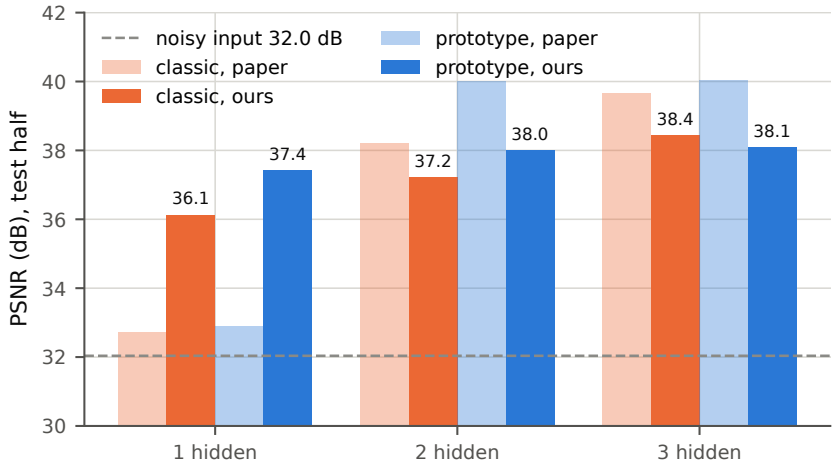


Figure 9: Paper Table I as a chart: PSNR on the test half for one to three hidden neurons, paper values (light) against ours (dark). Run `ecg/20260820T162014`.

5 The experiments under calibrated noise

5.1 Seven machines as noise models

For each of seven quantum computers we collected the vendor's published calibration numbers — two-qubit and one-qubit gate error, gate durations, T_1, T_2, readout error, and the qubit connectivity — into a device file whose every field carries the URL it was read from, the retrieval date (20 August 2026) and a verbatim quotation (Appendix B). Table 2 summarises them. Three things are worth knowing before reading any noise result. *Quantinuum publishes no gate durations and no T_1/T_2*; its idle error is instead charged from its published "memory error per depth-1 circuit time" (2.0×10^{-4} for H2-1, 2.2×10^{-4} for H1-1), and its simulated noise contains no idle error at all. *Rigetti publishes only a two-qubit error* for Ankaa-3 (and the 99.5 % headline is for a different gate than the universal one we need); every Rigetti number below is therefore an upper bound on the machine. *IQM's T_1/T_2 are a third-party snapshot*, not vendor data, and its T_2 measured without echo is 8.5 µs against the 21 µs echo value we use. Where a vendor publishes no number we model that channel as error-free and say so, rather than invent one.

5.2 Bit-count

The trained 11-qubit bit-count networks (prototype and classic) were evaluated exactly, as density matrices on the GPU, under each preset (run `bitcount_noise/20260820T145335Z`; Fig. 10). The model distribution moves by a total-variation distance of 0.005 (Quantinuum H2-1) to 0.039 (IQM Garnet); the KL to the target rises from 5.5×10^{-4} to 7×10^{-4} on H2-1 and to 6×10^{-3} on Garnet. The prototype network is better than the classic one under every preset, but the displacement *caused* by the noise ($D_{\mathrm{KL}}(\mathrm{clean} \,\|\, \mathrm{noisy})$, median 0.0021 vs 0.0022) is the same for both: both have identical two-qubit gate counts, and the prototype's advantage under noise is inherited from its better noiseless fit, not from any robustness of the neuron type. Scaling IBM Fez's noise by s (Fig. 11) gives $D_{\mathrm{KL}} \propto s^{1.7}$ and TV $\propto s^{0.95}$ — a linear response.

Can one train *under* the noise? On a 7-qubit version of the network (3 inputs, 3 hidden; the 11-qubit grid would have cost 13 hours and was not run) the answer is yes (Fig. 13): training with the noise model in the loop recovers the noisy KL by a median factor of 158 (prototype) and 361 (classic) — but the recovered parameters are *worse* on a noiseless simulator, i.e. the optimiser folds the channel into the angles rather than finding noise-robust ones.

5.3 Flappy Bird

The Flappy controller is a thresholded activation, so what matters is not how far $P(\mathrm{out} = 1)$ moves but whether it crosses $\frac{1}{2}$. Evaluated exactly under every preset on its 65×65 decision grid and then played for 20 held-out games (run `flappy_noise/20260820T154249Z`; Fig. 14, left), the controller *still plays perfectly on all seven machines*: only 1.5–4 % of the frames it actually visits are decided differently, because the noise contracts every activation toward $\frac{1}{2}$ by about the same amount (mean $|\Delta P| = 0.002$–0.018) and so rarely changes the ordering $P \gtrless \frac{1}{2}$. There is a factor of two of headroom: IBM Fez's noise scaled by 2 leaves 60 % of games perfect, by 4 none (Fig. 14, right).

Shot noise is a different matter (Fig. 15). Re-deciding every frame from N measurements of the output qubit, the same controller wins no game at $N = 1\,000$, 90 % at 4 000 and all at 10 000 on a noiseless simulator; on IBM Fez's model only 55 % at 4 000. The reason is the distribution of decision margins over the frames the agent plays: median $\delta = 0.019$, tenth percentile 0.004, first percentile 0.0002 (right panel). The measured flip rate per frame agrees with the average of $\Phi(-2\delta\sqrt{N})$ over that distribution to 3 %. A deterministic shift of the boundary (gate noise) is tolerated far past a stochastic one: Fez at scale 2 decides 20 % of frames differently and still

Table 2: The seven device presets (medians where the vendor publishes per-qubit tables). Two-qubit error ε_2, one-qubit error ε_1, readout error (mean of p_{01}, p_{10}), coherence times, two-qubit gate time. "—" = not published; the provenance of every entry is in Appendix B.

Machine	Connectivity	ε_2	ε_1	readout	T_1	T_2	2q gate (ns)
Quantinuum H1-1	all-to-all	9.7×10^{-4}	1.8×10^{-5}	2.3×10^{-3}	—	—	—
Quantinuum H2-1	all-to-all	1.1×10^{-3}	1.9×10^{-5}	1.0×10^{-3}	—	—	—
IonQ Forte	all-to-all	4.0×10^{-3}	2.0×10^{-4}	5.0×10^{-3}	55 s	1 s	950 000
Google Willow	square	4.1×10^{-3}	5.3×10^{-4}	8.0×10^{-3}	68 µs	89 µs	42
IBM Fez (Heron r2)	heavy-hex	3.9×10^{-3}	2.3×10^{-4}	7.7×10^{-3}	145 µs	88 µs	84
IQM Garnet	square	4.9×10^{-3}	8.0×10^{-4}	3.0×10^{-2}	27 µs	21 µs	30
Rigetti Ankaa-3	square	1.0×10^{-2}	—	—	—	—	72

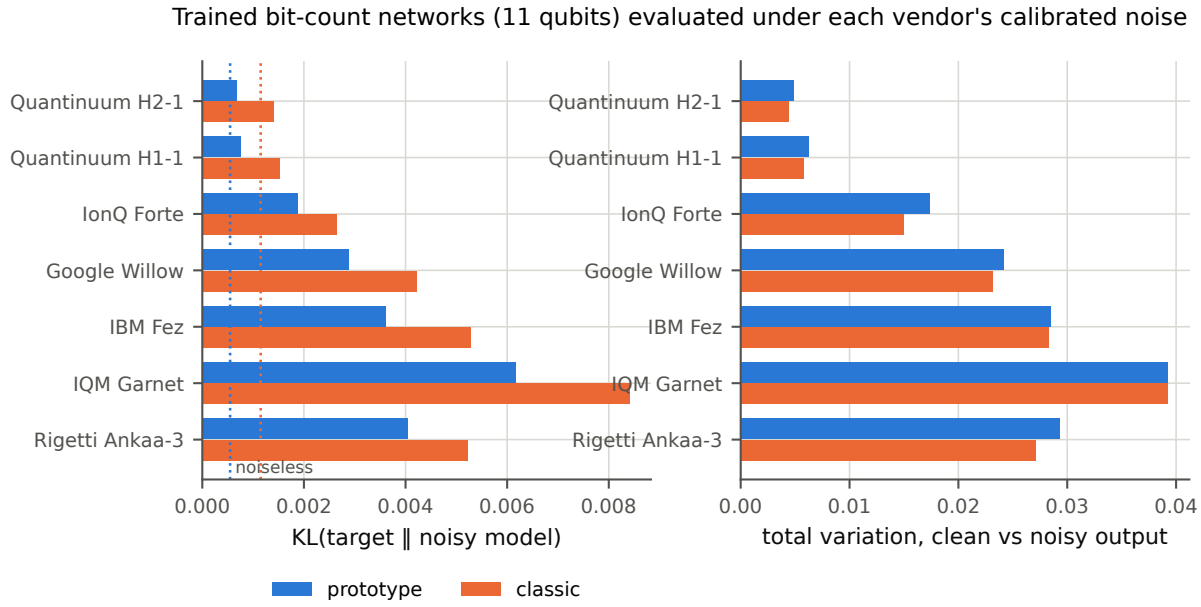


Figure 10: The trained bit-count networks under each machine's noise model. Left: KL to the target (dotted lines: the noiseless values). Right: how far the output distribution moved. Rigetti's row models only its two-qubit error. Run `bitcount_noise/20260820T145335Z`.

wins 60 % of games; 1 000 shots decide only 8.5 % differently and win none. A controller that is re-rolled every frame is a jittery controller, and two or three unlucky frames in a row are a collision. Fine-tuning under Fez's noise for five sweeps changes nothing (the training episodes already all survive, so the objective has no slope).

5.4 ECG and Iris (reduced run)

The ECG and Iris grids were run in a *reduced* configuration only (run `ecg_iris_noise/20260820T155618Z`): ECG on 8 test windows instead of 898 and two presets instead of seven; Iris with 16 Monte-Carlo trajectories per evaluation instead of 2 000 (per-outcome standard error $\approx 12\,\%$), two presets, and perfect preparation of the dataset state assumed. The measured cost of the full grids — 4–6 CPU-hours for ECG, 8 hours for Iris — put them outside the session's budget, and they were not run. What the reduced run shows (Fig. 16) is indicative only: ECG denoising loses −0.1 to +0.4 dB under Quantinuum H2-1 and 1.6–2.7 dB (classic) under IBM Fez, while the prototype cells *gain* up to 3.8 dB under Fez on eight windows — a small-sample artefact, not a finding. Iris keeps its 95 % accuracy under H2-1 and loses it under Fez ($64 \pm 44\,\%$ over two trajectory seeds) for the prototype network, whose read-out is a conditional distribution over 4 096 input

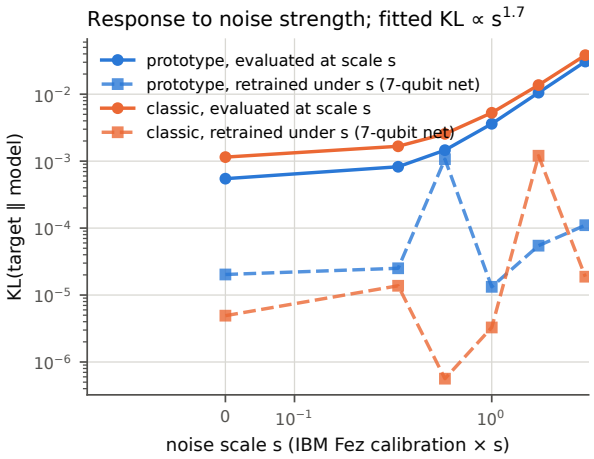


Figure 11: Response to noise strength: IBM Fez's calibration scaled by s (solid), and a 7-qubit network retrained with the scaled noise in the loop (dashed). Same run.

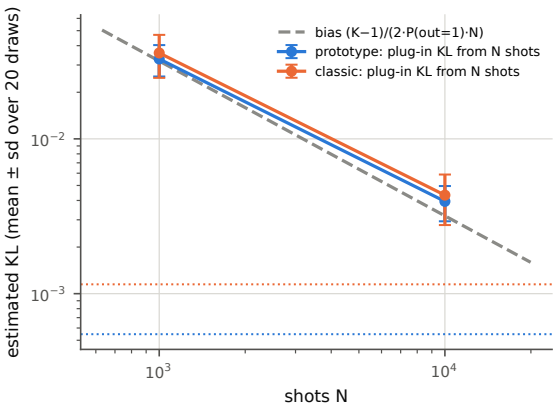


Figure 12: Estimating the KL from N shots: the plug-in estimate sits above the true value by $(K-1)/(2\,P(\text{out}=1)\,N)$, with $K=32$ bins and only $P(\text{out}=1)\approx 0.49$ of the shots usable.

strings and therefore directly exposed to probability leaking out of the 20-code support. One noisy sweep of ECG fine-tuning gains 2.7 dB under Fez and 3.7 dB noiselessly — it finishes the clean optimisation, it does not fold the channel in.

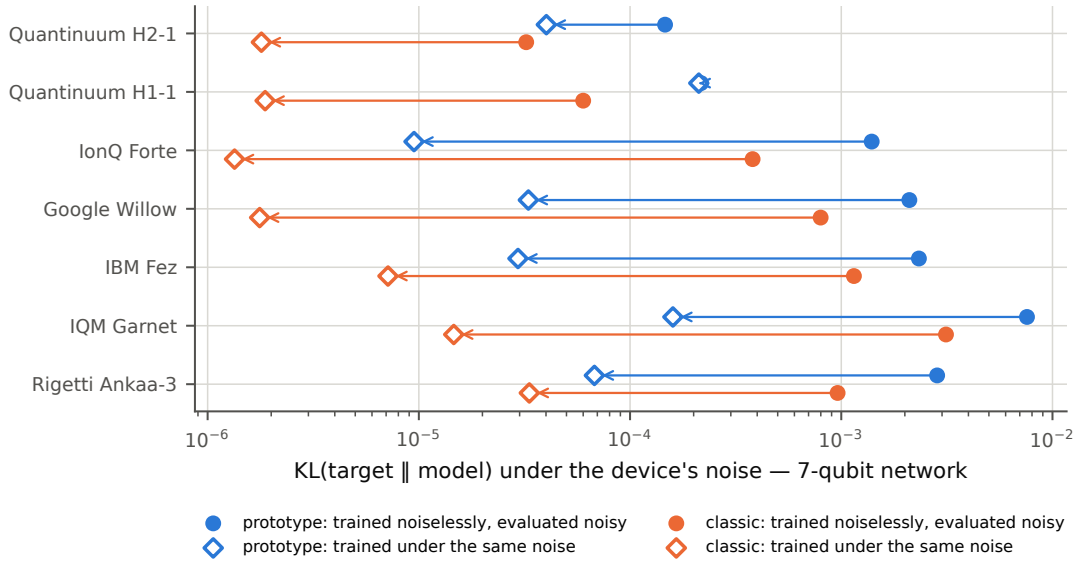


Figure 13: Noise-aware training (7-qubit bit-count network, three seeds): KL under each machine's noise for parameters trained noiselessly (filled) and for parameters trained with the same noise model in the loop (open), median of three seeds.

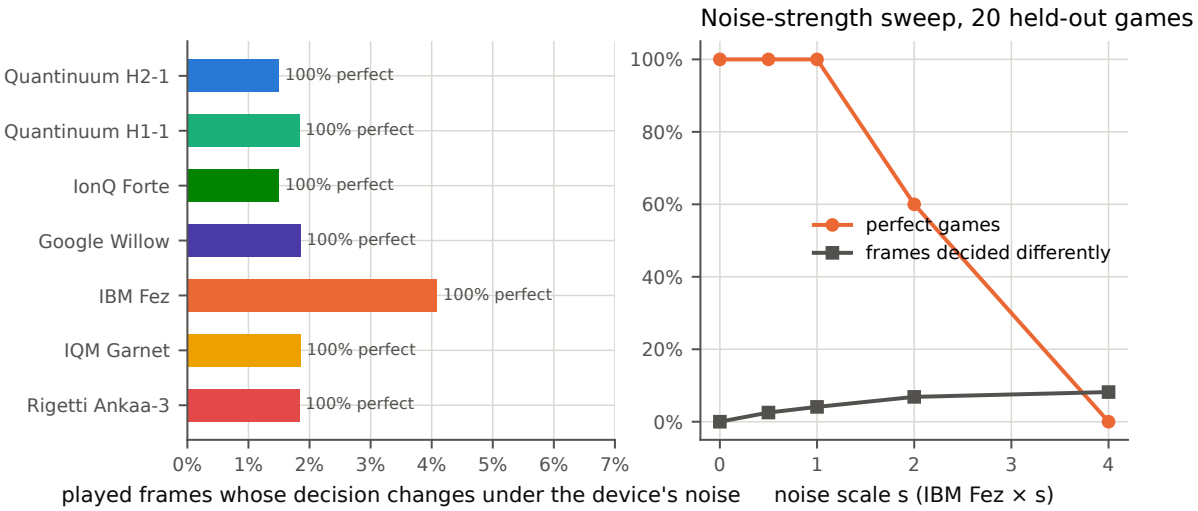


Figure 14: Flappy under gate noise. Left: fraction of played frames whose decision changes under each machine's model; every machine still yields 100 % perfect games. Right: IBM Fez's noise scaled by s. Run `flappy_noise/20260820T154249Z`.

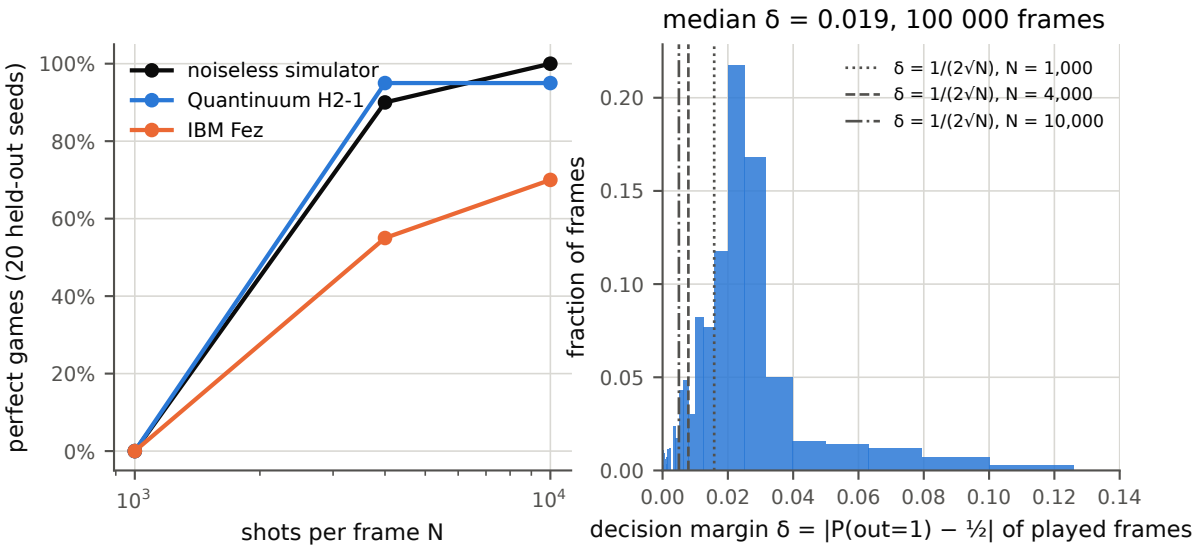


Figure 15: Flappy under shot noise. Left: perfect games when every frame is decided from N shots. Right: distribution of the decision margin δ over the 100 000 frames the noiseless controller plays; the vertical lines mark the margin that N shots can just resolve. Same run.

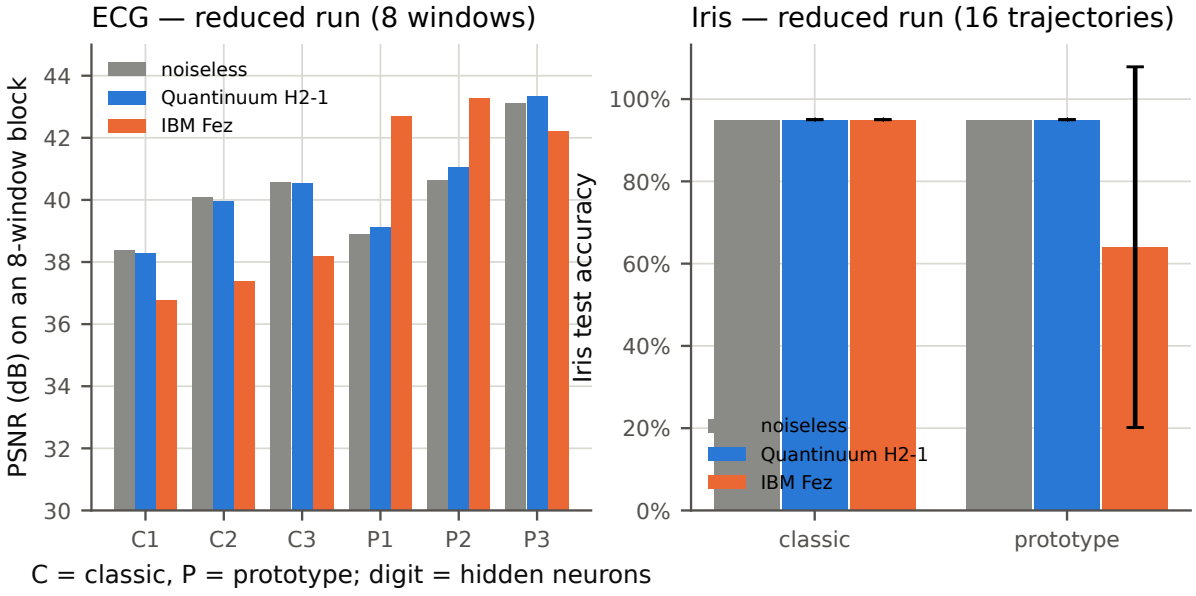


Figure 16: Reduced-configuration noise runs, quoted for the record. Left: ECG PSNR on an 8-window block. Right: Iris test accuracy from 16 trajectories (error bars: two trajectory seeds). Run `ecg_iris_noise/20260820T155618Z`.

6 What it takes to run this on a real machine

All numbers in this section are printed by `experiments/hardware_plan.py` into the run directory `hardware-plan/20260821T084554Z` (file `numbers.md`); the script joins every calibration number it uses to its provenance entry and refuses to run if one is missing.

6.1 Compilation: the prototype network is the worst case for a lattice

Every prototype neuron touches every input, so the network's two-qubit gates form a complete bipartite graph between inputs and hidden qubits. On an all-to-all machine (trapped ions) each controlled rotation costs exactly two native two-qubit gates and nothing else: 24 for Flappy, 60 for bit-count. On a lattice the compiler must insert SWAPs, three native gates each, to bring non-neighbouring qubits together. With our own router (greedy, no look-ahead) the bit-count circuit needs 327 native two-qubit gates on IBM's heavy-hex lattice and 207 on a square lattice, of which 82 % and 71 % are SWAP-borne (Fig. 17). IBM's own compiler does much better — 97 gates at its optimisation level 2, 3.4× fewer — so our router is a device-independent comparison instrument, not what one would submit. Flappy's circuit is 12.9 µs long on Fez (1.56 µs of which is readout) and 17.9 ms on IonQ Forte, whose gates take a millisecond.

6.2 Expected fidelity

Figure 18 stacks the four factors of Eq. (7) for the two smallest circuits, with the exactly simulated total-variation distance between the compiled circuit run clean and run under the machine's noise model next to each bar. Flappy at 7 qubits is a high-fidelity circuit on the trapped-ion machines ($F \approx 0.92$ on H2-1, 0.83 on Forte), marginal on Willow and Fez (0.63; 0.46 with our router, 0.75 with IBM's compiler), and poor on Garnet (0.24). Bit-count at 11 qubits is feasible on H-series (0.81) and Forte (0.64), usable with IBM's compiler on Fez (0.48), and dead on Garnet (0.008), whose routed circuit lasts 86 % of its T_2. Idling is charged only over each qubit's gaps in the schedule (relaxation during a gate is already inside the vendor's gate error); the older variant that charged the whole circuit duration on every qubit was 10–15 points more pessimistic on the superconducting machines and is printed beside it in the run directory. Two cautions. The simulated distances of the *compiled* circuits (0.03 on H-series, 0.16 on Forte, 0.43 on Fez) are 5–15× larger than those of the uncompiled circuits in Section 5.2: two native gates per controlled rotation, resynthesised one-qubit gates and routing all cost. And the vendors' two-qubit errors are measured in isolation; IBM's own layered benchmark for Fez is 1.7× worse than its isolated gate error, so the lattice numbers are optimistic.

6.3 Shots, time and money

The shot budgets follow from Section 2. An activation to ±0.005 needs 10 000 shots. The 32-bin bit-count histogram is estimated from the shots in which the output fired ($P(\text{out}=1) = 0.487$ of them), and the plug-in KL is then biased upward by $(K-1)/(2 \cdot 0.487 \cdot N) = 31.8/N$; reaching the trained model's 5.5×10^{-4} needs 58 000 shots, which we round to a 64 000-shot job. The ECG denoiser's PSNR is capped by shot noise at $10 \log_{10}(4N)$: 2 500 shots per sample cap it at exactly 40 dB, the paper's Table I value, and a 1 800-sample trace at 25 000 shots is 4.5×10^7 shots — ECG is a simulator experiment. Figure 19 converts the circuit times to machine time per job: milliseconds to two seconds on the superconducting machines, where the per-job overhead dominates, but 72 s (Flappy) and 32 min (bit-count) on IonQ Forte, where the circuit time does. Prices (retrieved 20 August 2026; Table 3): a 4 000-shot Flappy job costs about $320 on Forte through Braket, $6 on Garnet, 323 Quantinuum credits (≈ $4 000 at the Azure plan rate), and is free on IBM's Open plan. Training on hardware is out of the question — the paper's 2 160

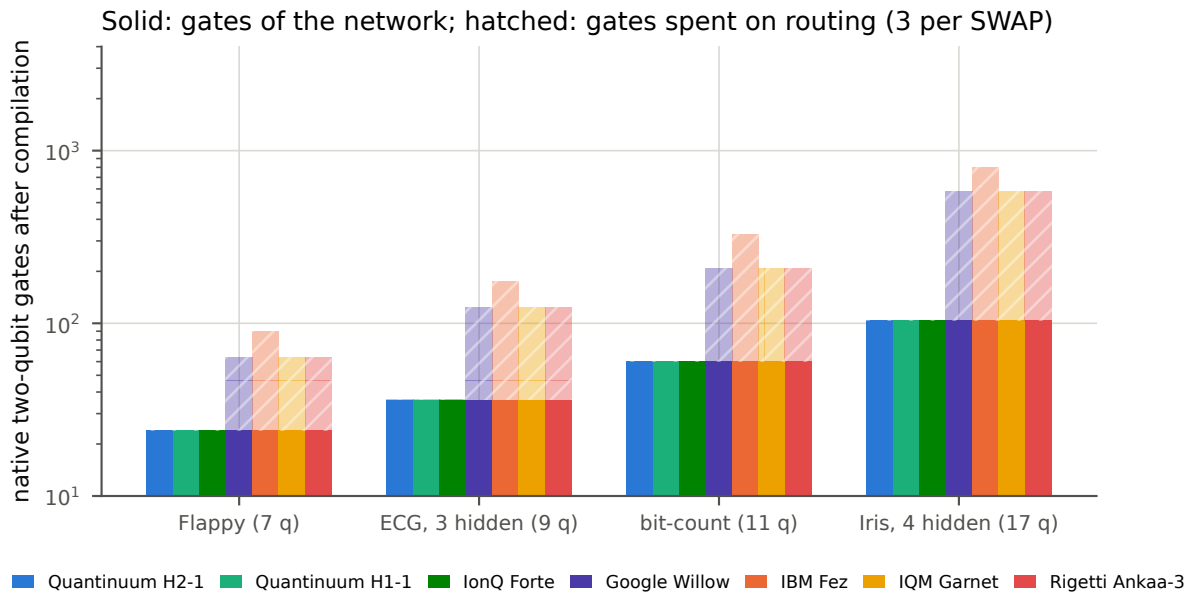


Figure 17: Native two-qubit gates after compiling each network for each machine (our router; log scale). The all-to-all machines need only the network's own gates; the lattices spend most of their budget on routing. Run `hardware-plan/20260821T084554Z`.

Table 3: Cost per job (numbers.md §cost). IBM: \$1.60 per QPU second plus a 30 s per-job overhead (both assumptions, Appendix A); Quantinuum: the vendor's published credit formula, converted at the Azure plan rate; Braket: published per-shot prices.

Machine	model	Flappy, 4 000 shots	bit-count, 64 000 shots
IBM Fez	\$1.60/s; 52 ms resp. 1.8 s of QPU	\$0.08 (\$48 with overhead)	\$2.85 (\$51)
IBM Open plan	10 min per 28 days, free	\$0	\$0
Quantinuum H1/H2	credits $= 5+(N_1+10N_2+5N_m)C/5000$	323 credits ≈ \$4 030	11 920 credits ≈ \$149 000
IonQ Forte (Braket)	\$0.30 + \$0.08/shot	\$320	\$5 120
IQM Garnet (Braket)	\$0.30 + \$0.00145/shot	\$6.10	\$93

evaluations at 64 000 shots are 1.4×10^8 shots, 48 days of Forte time or \$11 M — but fine-tuning the six output angles (90 evaluations at 4 000 shots) is 1.8 hours on Forte and about 45 minutes on Fez. Live Flappy play is equally out (1 500 frames × 4 000 shots × 17.9 ms = 30 hours per episode on Forte); the falsifiable hardware observable is the activation map of Fig. 7 on a grid of observations, 16 jobs.

6.4 Recommendation

Run the 7-qubit Flappy activation map first: on IBM's free plan to debug the pipeline (readout calibration, then zero-noise extrapolation by gate folding, both validated offline against the noise model first), then on IonQ Forte for the confirmation (\$1 300 for a 4×4 grid at 1 000 shots), and on Quantinuum H2-1 for the headline number if an allocation exists. The bit-count histogram is feasible on H-series and Forte as a *shape* demonstration but not as a KL comparison (device error puts the KL floor at 10^{-2}–10^{-1}, two orders above the simulated 5.5×10^{-4}). Whether Iris is within reach cannot be settled yet: what our runs prepare is a uniform superposition over the 12–14 *distinct* training codes (not the 120 samples an earlier draft assumed), and the sparse construction of [10] has leading term $k \cdot n = 144$–168 two-qubit gates, a fidelity of ≈ 0.83 on H2-1

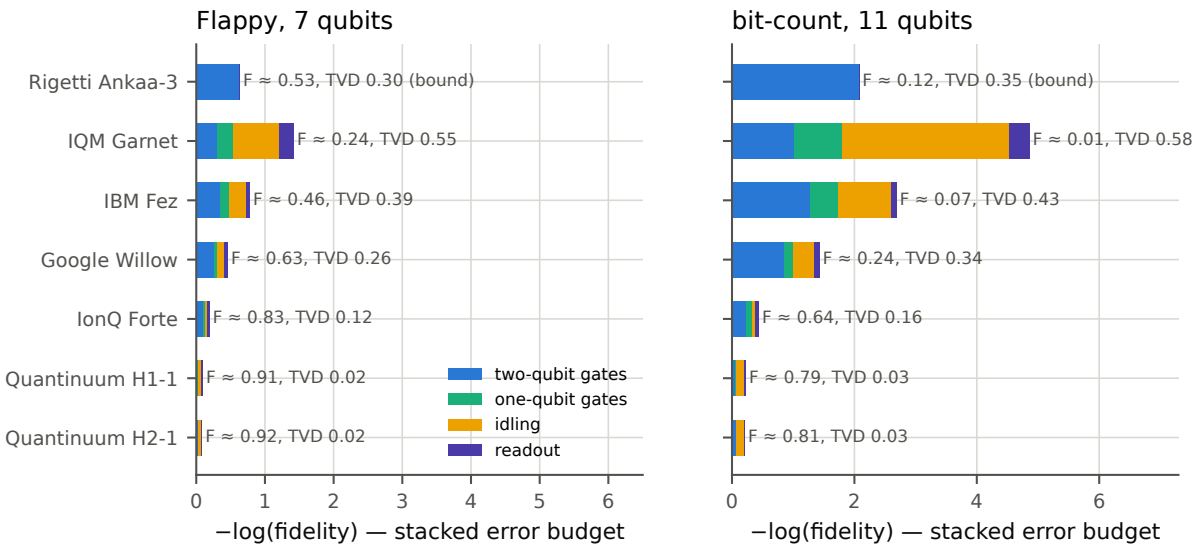


Figure 18: Error budget per machine for the Flappy (left) and bit-count (right) circuits: the stacked bars are $-\log$ of each factor of Eq. (7), so their total is $-\log F$; the labels give F and the simulated total-variation distance of the compiled circuit. Quantinuum's idle term comes from its published memory error; Rigetti's is a bound (two-qubit error only).

and ≈ 0.75 with the network — but its constant is unknown, and the dense alternatives (8 168 gates [8], $\approx$ 3 925 [9]) are hopeless. An exact answer needs an actual sparse state-preparation circuit, which this repository does not build. Four things the code still lacks for a live submission are listed in the repository's hardware plan (credentials, the vendors' own routing, asynchronous job handles, readout-unfolding and folding utilities).

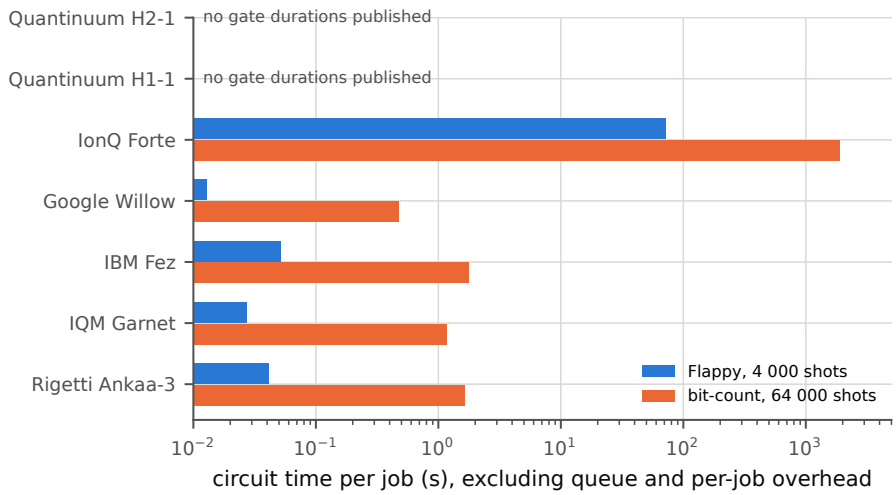


Figure 19: Machine time per job, from the compiled circuit's duration (our router; readout included where published). Quantinuum publishes no durations.

7 The shot-noise programme

Section 5 found that, at today's published error rates, what limits a hardware demonstration of these networks is not gate error but the number of measurements. Two mechanisms were identified: a thresholded controller whose decisions sit close to its threshold flips at a rate set by $\Phi(-2\delta\sqrt{N})$; and a histogram loss estimated from N shots is biased upward by $(K-1)/(2\,P(\text{out}=1)\,N)$ and discards the half of the shots in which the output did not fire. Six remedies were proposed; this section reports the three we implemented and tested (Table 4). All runs are noiseless state-vector simulations plus sampling, minutes each.

7.1 Margin-aware training (negative result)

The idea: if the controller could be trained so that every frame it visits has $\delta \geq 0.1$ instead of the median 0.019, the shots needed per frame would fall by $(0.1/0.019)^2 \approx 25\times$, at no cost on a noiseless simulator. We added to the training objective a penalty of $w \times 0.01 \times \max(0, 1 - \delta/m)$ per played frame — a fraction $w < 1$ of the 0.01 reward for surviving that frame, so that surviving a frame is always worth more than dying confidently — and fine-tuned the trained controller for 25 sweeps with $m = 0.1$ and $w \in \{0.25, 0.5, 1\}$ (run `flappy_shots/20260821T084601Z`). A first attempt with a *mean* hinge of weight 20, trained from scratch, had collapsed to a policy that died in 30 frames with perfectly confident decisions (run `…T075048Z`): any penalty that can outweigh the per-frame reward selects for dying early, which is why the per-frame form is necessary.

Figure 20 (left) shows the outcome: the margin quantiles hardly move (median 0.019 → 0.021–0.024, tenth percentile 0.004 → 0.006). All three fine-tuned controllers still play perfectly on exact probabilities, but under shot noise they play *worse* than the original at $w \geq 0.5$ (50 % and 80 % perfect games at 10 000 shots against 100 %). The explanation is in Fig. 7: the activation is a smooth surface over the two observations, and the near-boundary frames are those in which the bird hovers at the boundary — a property of the game's dynamics, not of the surface's steepness. Penalising them moves the boundary rather than steepening it, and a moved boundary is a less stable controller. A 17-parameter network has no spare capacity to steepen its surface. We record the remedy as not working for this network.

Table 4: The shot-noise programme: what was tried and what came out. Items 4–6 (joint-distribution loss or amplitude amplification to use all shots; bias-corrected KL estimators; amplitude estimation) are designed but not implemented.

Remedy	Mechanism	Outcome
1. Margin-aware training	add a penalty for frames whose margin δ is below $m = 0.1$	**negative**: margins of this network barely move; play under shot noise gets worse
2. Common random numbers	evaluate the two finite-difference probes on the same measurement randomness (maximal coupling)	derivative variance ÷3 at our reproduction's step ($c = 0.2$), ÷12–22 at small steps; sampled training improves at $N \leq 4\,000$ but stays 10× above the exact-probability result, consistent with the bias of the loss itself
3. Sequential decisions	Wald's sequential test on the sign of $P(\text{out} = 1) - \frac{1}{2}$ per frame	**positive**: 100 % perfect games at about 2 200 shots/frame, 4.6× fewer than the fixed budget that achieves the same



Figure 20: Flappy under shot noise, remedies 1 and 3. Left: decision-margin quantiles of the frames played by the original and the three margin-aware controllers. Middle: perfect games against shots per frame for a fixed budget (lines) and for the sequential test (stars, at the mean shots it actually spent). Right: mean shots per frame of the sequential test against its indifference margin δ_0. Run `flappy_shots/20260821T084601Z`.

7.2 Common random numbers for the finite-difference optimiser

Mechanism. The paper's optimiser estimates a slope as $(f(x+c) - f(x-c))/2c$. If each probe is estimated from N independent shots with variance σ^2, the slope has variance $\sigma^2/2c^2$ — at our reproduction's $c = 0.2$ that is $12.5\sigma^2$, which is why finite differences on a sampling backend stall. If the two probes are instead computed from the *same* random draws, the noise largely cancels. The standard way (feeding the same uniform random numbers to the inverse cumulative distribution of both probes) turned out to give only a 1.3× reduction here, because a parameter change moves probability between distant bit strings and so shifts every later bin boundary. What works is the *maximal coupling* [6]: sample the first probe, then for the second keep each shot's outcome i with probability $\min(1, p_{\text{new}}(i)/p_{\text{ref}}(i))$ and redraw it otherwise from the excess $(p_{\text{new}} - p_{\text{ref}})_+$. The second sample has exactly the right distribution, and only $N \cdot \text{TV}(p_{\text{new}}, p_{\text{ref}})$ shots change — the fewest any coupling can achieve. Since TV $\propto c$, the difference of the two estimates has variance $O(c/N)$ and the slope, after division by $2c$, has variance $O(1/(cN))$ instead of $O(1/(c^2N))$: a gain of order $1/c$. Coupling reduces the variance of the slope; it does not remove the bias of a non-linear loss such as the plug-in KL. On hardware the analogue is to interleave the $\pm c$ circuits in one job so they share the same drift.

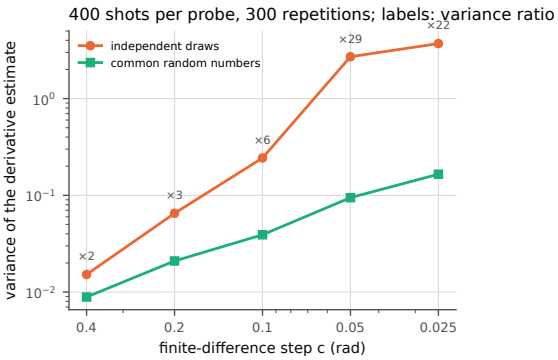


Figure 21: Variance of the finite-difference slope of the bit-count loss at 400 shots per probe, with independent draws and with maximally coupled draws; labels give the ratio.

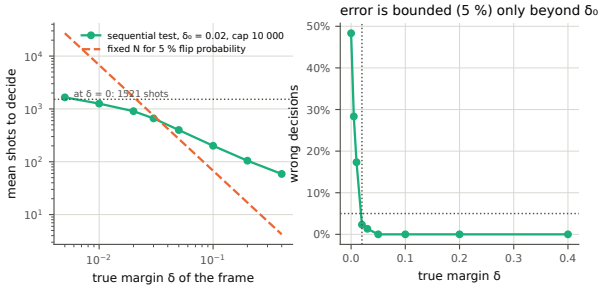


Figure 22: The sequential test ($\delta_0 = 0.02$, 5 % error bounds, cap 10 000): mean shots to decide a frame against its true margin (left), and its error rate (right), which is bounded only beyond δ_0.

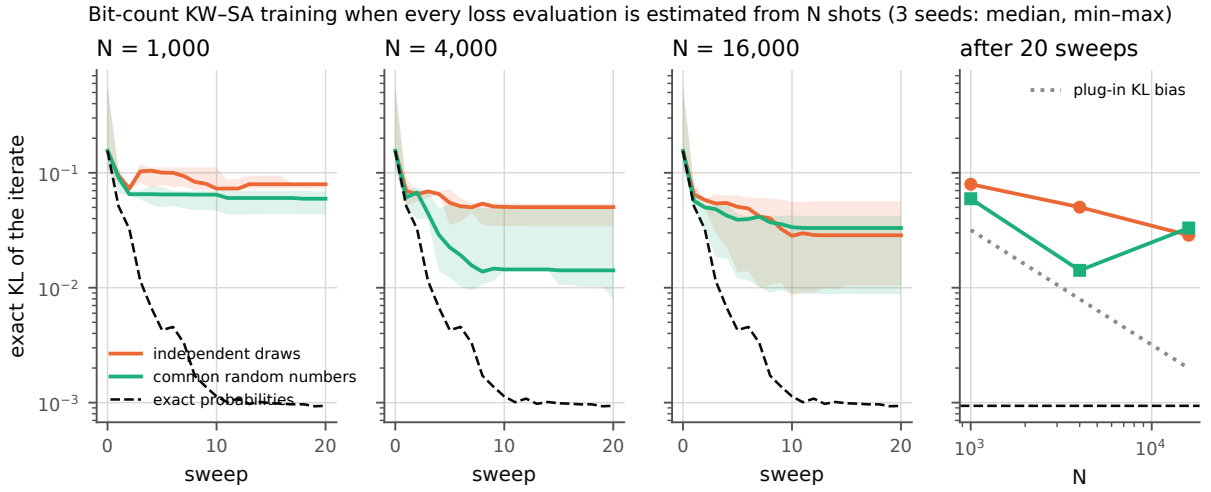


Figure 23: Bit-count training when every loss evaluation is estimated from N shots (three seeds; median and range), with and without common random numbers, against training on exact probabilities. Right: the exact KL reached after 20 sweeps against N, with the plug-in bias of the loss for comparison. Run `bitcount_shots/20260821T084558Z`.

Measurement. Figure 21 measures the variance of one slope of the bit-count loss at 400 shots per probe: the reduction is ×2–3 at $c = 0.2$–0.4 and ×22–29 at $c = 0.025$–0.05, as the $1/c$ scaling predicts, and both estimates are unbiased for the exact slope. Figure 23 then trains the 11-qubit bit-count network with our reproduction's schedule (the paper publishes none) on a sampling backend (run `bitcount_shots/20260821T084558Z`): at $N = 1\,000$ shots per evaluation the exact KL after 20 sweeps improves from 0.080 to 0.059 with common random numbers, at 4 000 from 0.050 to 0.014, and at 16 000 the two are within the seed-to-seed spread (0.029 vs 0.033). All of them remain an order of magnitude above the exact-probability result (0.0009). The right panel suggests why: the *loss* is the plug-in KL of a 32-bin histogram from about $N/2$ usable shots, biased by $31.8/N$, and the reached KL tracks one to three times that bias. This is consistent with the estimator, not the optimiser, being what now limits sampled training, but three seeds (one of which stalls at 0.014 even on exact probabilities) cannot separate the two; a bias-corrected or joint loss under paired initialisations is the test, and it has not been run. Our step schedule ($c_0 = 0.2$ decaying only to 0.15 over 20 sweeps) also keeps the optimiser in the regime where the reduction is smallest.

Table 5: Flappy, original controller, 20 held-out games: fixed shot budgets against the sequential test ($\alpha = \beta = 0.05$). "Flips" are frames decided differently from the exact rule. Run `flappy_shots/20260821T084601Z`.

Decision rule	perfect games (%)	median pipes	flips (%)	mean shots/frame
fixed $N = 1\,000$	5	23	8.36	1000
fixed $N = 2\,000$	70	110	7.40	2000
fixed $N = 4\,000$	85	110	6.09	4000
fixed $N = 10\,000$	100	110	4.47	10 000
sequential, $\delta_0 = 0.02$, cap 10 000	30	88	7.19	822
sequential, $\delta_0 = 0.01$, cap 4 000	80	110	6.12	1790
sequential, $\delta_0 = 0.005$, cap 4 000	85	110	5.98	2916
sequential, $\delta_0 = 0.01$, cap 10 000	100	110	5.06	2191
sequential, $\delta_0 = 0.005$, cap 10 000	100	110	4.50	4358

7.3 Sequential decisions (positive result)

Mechanism. A fixed budget of N shots per frame wastes most of them: a frame with $\delta = 0.3$ is decided correctly after a dozen shots. Wald's sequential probability ratio test [5] between the hypotheses "$P(\text{out} = 1) = \frac{1}{2} + \delta_0$" and "$P(\text{out} = 1) = \frac{1}{2} - \delta_0$" draws shots in blocks, after n shots with k ones computes the log-likelihood ratio $\Lambda = (2k - n) \log \frac{1+2\delta_0}{1-2\delta_0}$, and stops as soon as $\Lambda \geq \log \frac{1-\beta}{\alpha}$ (flap) or $\Lambda \leq \log \frac{\beta}{1-\alpha}$ (do not flap), up to a cap $N_{\max}$ after which the majority decides. For a frame whose true margin exceeds δ_0 the error probability is at most about α ($= \beta = 5\,\%$ here); for a frame nearer the boundary than δ_0 the test is slow and its error is not bounded (Fig. 22).

Measurement. Played on the same 20 held-out games (Fig. 20, middle and right; Table 5), the sequential rule with $\delta_0 = 0.01$ and cap 10 000 wins *every* game at a mean of 2 191 shots per frame — the fixed budget needs 10 000 for the same result, 4.6× more — and with $\delta_0 = 0.005$ and cap 4 000 it matches the fixed 4 000-shot result (85 %) at 2 916. Setting δ_0 at the median margin (0.02) fails (30 % perfect at 820 shots): the frames that decide a game are the near-boundary ones, and the indifference margin must sit below *their* margins (the tenth percentile, 0.004–0.006), not the typical one. The rule needs no change to the network, no retraining, and is what one would submit to a machine that supports mid-circuit decisions or, more simply, adaptive batch sizes per frame.

8 What we learned

1. **The paper's central claims reproduce.** The prototype neuron's activation surface, the bit-count histogram and its convergence, the Iris overfitting contrast, perfect Flappy play (on our clone) and the ECG denoising numbers (within 2 dB on a different signal) all come out of a clean-room implementation. What does not: the iterative baseline of Fig. 6 (ours stalls at $D_{\text{KL}} \approx 0.15$) and the weight picture of Fig. 9. Both are under-specified in the paper.
2. **Reading figures is part of reproduction.** The bit-count target ($(\text{ones} + 1)/112$), the Iris label weighting ($(\ell + 1)^2$) and the angle encoding ($\theta = 2 \arcsin \sqrt{s}$, from a printed gate angle in the companion paper) all had to be read off figures; each was then confirmed to the figure's precision. Where a figure was over-read — the Iris split ratio "justified" by bar counts — an independent review caught it and the justification was withdrawn (Section 4.2).
3. **The prototype network's advantage under noise is inherited, not intrinsic.** Prototype and classic networks have the same two-qubit gate count and the noise displaces both by the same amount; the prototype wins under noise because it fits better without noise.

4. **Gate noise at published levels is not the obstacle for these networks; shots are.** Every 7–11-qubit network survives every vendor's calibrated noise model with a qualitatively intact result. The Flappy controller needs 10^4 shots per frame with a fixed budget; the bit-count loss is biased by $31.8/N$. The one remedy that pays now is a sequential measurement rule (4.6× fewer shots, no retraining); margin-aware training does not work for a 17-parameter network; common random numbers reduce the optimiser's noise by $1/c$ but leave the estimator's bias, which is the likely (untested) limit of sampled training.

5. **Compiled circuits are 5–15× more noise-sensitive than logical ones.** Two native gates per controlled rotation, resynthesised single-qubit gates and routing on a lattice all cost; a complete-bipartite network is the worst case for a lattice, and the vendor's own compiler must be used for any superconducting submission. Trapped ions are the natural home of this architecture; their price is circuit time (milliseconds per shot).

6. **An optimiser that keeps a stale incumbent turns a harmless change into a negative result.** The "rotating seeds break training" finding of the first session was an artefact of comparing losses measured on different episode batches; re-scoring the incumbent restores 25/25 perfect games. Any non-stationary objective (mini-batches, hardware drift) needs this.

7. **Penalties must be priced in the currency of the reward.** A margin hinge that could outweigh the per-frame survival reward produced a confidently dead controller; the per-frame form below the per-frame reward cannot. The same arithmetic applies to any shaping term.

8. **Provenance has to be mechanical.** The first hardware plan contained a dozen wrong numbers — a state-preparation gate count off by a factor of two, a sentence with its inequality reversed, costs computed by hand. The fix was a script that emits every number and refuses to run if any calibration input lacks a source; the report now quotes only that script.

Not done, and why. Noise-aware training of the 11-qubit bit-count network (13 hours for its grid) and the full ECG/Iris noise grids (4–8 hours) exceeded the session's compute ceiling and were not run; their per-cell costs are recorded so that a future session can decide. Items 4–6 of the shot-noise programme — a loss that uses all shots, bias-corrected estimators, and amplitude estimation — are designed but not implemented; item 4 is the natural next step, since the estimator bias is what now limits sampled training. Nothing has been submitted to a quantum computer; the plan of Section 6.4 is ready to be.

A Choices the paper leaves open

Each entry is marked `[ASSUMPTION]` at the place in the code where it enters.

Optimiser schedule (Section 3.2).
Step sizes $a_n = a_0/(n+1)^{0.602}$, difference step $c_n = c_0/(n+1)^{0.101}$, geometric cooling; one iteration is a coordinate sweep. Bit-count: $a_0 = 20$, $c_0 = 0.2$, $T_0 = 0.02$, cooling 0.7, 20 sweeps. Flappy: $a_0 = 0.4$, $c_0 = 0.25$, $T_0 = 0.05$, cooling 0.9, 25 sweeps after 300 random restarts. Iris: $a_0 = 2$, decay exponent 0.3, $c_0 = 0.1$, 40 sweeps. ECG: 100 sweeps with restarts.

Iterative baseline (Section 4.1).
One optimiser sweep on the squared error of one random input per iteration; the paper does not describe its scheme.

Angle encoding (Eq. (3)).
$\theta = 2 \arcsin \sqrt{s}$, pinned by a printed angle in [2]; $\theta = \pi s$ tested as an ablation.

KL regularisation.
The model probability is floored at 10^{-12} inside the logarithm.

Iris (Section 4.2).
Stratified one-third training split; flowers sharing a code are merged and their labels averaged; the KL is normalised over all 4 096 input strings (restricting it to the dataset's codes lets the network exploit interference between examples and reach a training loss of 10^{-5} while reading out at chance); label weight $\propto (\ell+1)^2$ read from Fig. 8; labels predicted by nearest rounding of the read-out weight.

Flappy (Section 4.3).
Our own clone: 288×512 screen, flap impulse −7, gap 120, pipe spacing 180, gravity 1 (the classic constants are unplayable by a memoryless two-feature policy); reward 0.01 per frame plus 1 per pipe, tracking term |height − gap| with weight 1; 5 training episodes of 1 500 frames, 20 held-out seeds 5–24, 5 000-frame cap (110 pipes); decisions cached on a 65×65 grid of observations.

ECG (Section 4.4).
Synthetic McSharry-form signal tuned to Fig. 13 and MIT-BIH record 100; Gaussian noise $\sigma = 0.02$; mirror-padded edges; first half train, second half test; PSNR with $R =$ range of the clean signal; SSIM in a 7-sample sliding window.

Noise model (Section 5.1).
Depolarising at the vendor's average gate error, relaxation during gates and as-soon-as-possible idling, readout confusion; median values where vendors publish per-qubit tables; no crosstalk (no vendor publishes a usable number); where a vendor publishes nothing for a channel it is modelled as error-free and flagged. Quantinuum's idle error is charged as $(1-m)^{n_{\text{qubits}}\cdot\text{depth}}$ from its memory error per depth-1 time; IQM's and Rigetti's unpublished one-qubit gate durations are set to 40 ns.

Fidelity estimate (Eq. (7)).
First-order product of per-gate survival probabilities; readout factor $(1-(p_{01}+p_{10})/2)^n$.

Costs (Section 6.3).
IBM $1.60 per QPU second (2022 launch figure, corroborated by a secondary $96/min report) plus 30 s per-job overhead; Quantinuum $12.5 per credit (Azure Standard plan divided by its allotment); Braket and Azure per-shot prices as retrieved on 20 August 2026; Quantinuum's credit formula with $N_m = 2\times$ qubits.

State preparation (Section 6.4).
Dense counts $2^{n+1} - 2n$ [8] and $\approx \frac{23}{24}2^n$ [9]; sparse *leading term* $k \cdot n$ of an $O(kn)$ construction [10] with k the 12–14 distinct training codes actually prepared (constant unknown).

Compilation.
Lattice machines are compiled onto a synthetic heavy-hex or square patch of the circuit's size with our greedy router; Google's and Rigetti's gate sets are rebased through a generic fallback. Idle relaxation is charged over each qubit's schedule gaps only.

Shot-noise programme (Section 7).
Margin penalty $w \cdot 0.01 \cdot \max(0, 1-\delta/m)$ per frame with $m = 0.1$; sequential test with $\alpha = \beta = 0.05$, blocks of 20 shots; common random numbers by maximal coupling to the first probe of each coordinate.

B Provenance of the calibration data

Every calibration number used in Sections 5 and 6 is stored in a device file (`src/ptqnn/noise/devices/*.toml`) together with a `[[provenance]]` block per field: the source URL, the kind of source (vendor calibration page, vendor paper, third-party calibration snapshot), the retrieval date and a verbatim

quotation of the sentence or table row the number was read from. The hardware-plan script joins these blocks to every field it reads and fails if any is missing; the full join (one row per device and field, with quotations) is the file `numbers.md` of run `hardware-plan/20260821T084554Z`, section "provenance". The sources, in brief:

IBM Fez (Heron r2).
One coherent calibration snapshot, backend version 1.2.8 of 2025-02-26, shipped as `props_fez.json` inside `qiskit-ibm-runtime`'s fake provider; medians over 156 qubits and 352 directed CZ edges computed by us from that table. The file records the same chip drifting between CZ error 2.85×10^{-3} (2024-07) and 3.90×10^{-3} (2025-02).

Quantinuum H1-1, H2-1.
The vendor's performance-validation page (https://docs.quantinuum.com/systems/user_guide/hardware_user_guide/performance_validation.html): mean two-qubit and one-qubit infidelities, SPAM error, and "Memory Error per Depth-1 Circuit Time"; no gate durations and no T_1/T_2 are published.

IonQ Forte.
The vendor's published specification (average gate fidelities, gate durations, $T_1 = 55\,\mathrm{s}$, $T_2 = 1\,\mathrm{s}$).

Google Willow.
The supplementary information of the Willow paper (*Nature*, 2024): single- and two-qubit gate errors, coherence times, cycle time. The machine is not publicly accessible; the preset exists for simulation only.

IQM Garnet.
Vendor data for gate errors and readout; T_1 and T_2 (echo) from a third-party calibration snapshot (two epochs recorded, December 2025 and March 2026), with the caveat that T_2 without echo is $\approx 8.5\,\mu\mathrm{s}$.

Rigetti Ankaa-3.
Only the two-qubit (iSWAP) error 10^{-2} is published for the universal gate; the 99.5 % headline refers to the fSim gate. No one-qubit error, readout error, T_1 or T_2.

The per-field table is not reproduced here — it runs to some sixty rows with quotations — but it is the ground truth the numbers in this document rest on, and it is a single command away: `uv run python experiments/hardware_plan.py`.

References

[1] B. Rosenhahn and C. Hirche, *Prototype Learning for Quantum Neural Networks*, manuscript `QPrototypeLearning-5.pdf` (2026). Figure and equation numbers cited as "Fig. 5", "Eq. (4)" refer to this manuscript.

[2] B. Rosenhahn, T. J. Osborne and C. Hirche, *Stochastic Neural Networks for Quantum Devices*, arXiv:2602.22241 (2026).

[3] J. Kiefer and J. Wolfowitz, Stochastic estimation of the maximum of a regression function, *Ann. Math. Stat.* **23**, 462 (1952).

[4] J. C. Spall, *Introduction to Stochastic Search and Optimization*, Wiley (2003), §6 (gain-sequence exponents 0.602 and 0.101).

[5] A. Wald, Sequential tests of statistical hypotheses, *Ann. Math. Stat.* **16**, 117 (1945).

[6] T. Lindvall, *Lectures on the Coupling Method*, Wiley (1992), §I.2 (the maximal coupling).

[7] M. A. Nielsen, A simple formula for the average gate fidelity of a quantum dynamical operation, *Phys. Lett. A* **303**, 249 (2002).

[8] V. V. Shende, S. S. Bullock and I. L. Markov, Synthesis of quantum-logic circuits, *IEEE Trans. CAD* **25**, 1000 (2006).

[9] M. Plesch and Č. Brukner, Quantum-state preparation with universal gate decompositions, *Phys. Rev. A* **83**, 032302 (2011).

[10] N. Gleinig and T. Hoefler, An efficient algorithm for sparse quantum state preparation, *Proc. DAC* (2021).

[11] Z. Wang, A. C. Bovik, H. R. Sheikh and E. P. Simoncelli, Image quality assessment: from error visibility to structural similarity, *IEEE Trans. Image Process.* **13**, 600 (2004).

[12] P. E. McSharry, G. D. Clifford, L. Tarassenko and L. A. Smith, A dynamical model for generating synthetic electrocardiogram signals, *IEEE Trans. Biomed. Eng.* **50**, 289 (2003).